

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 February 2026
Team ID	LTVIP2026TMIDS81316
Project Name	TransLingua:AI-Powered Multi-Language Translator
Maximum Marks	4 Marks

Technical Architecture:

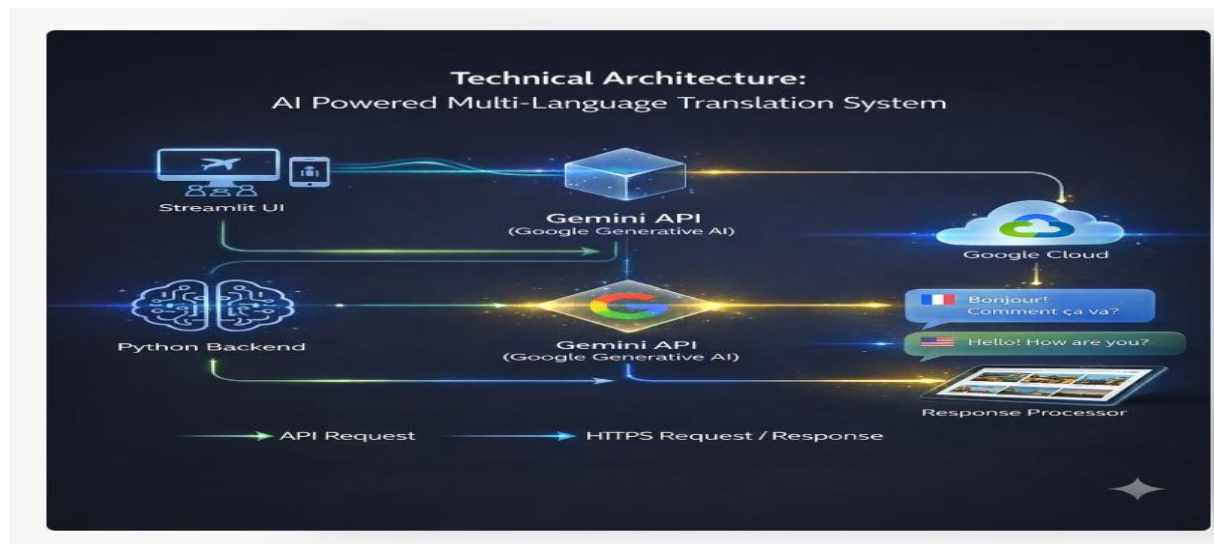


Table-1 : Components & Technologies:

S.No	Component	Description	Technology Used
1	User Interface	Interactive web interface where users enter text, select source and target languages, and view translated output	Streamlit
2	Application Logic	Handles user input, language detection, prompt construction, and API request handling	Python
3	Generative AI Engine	Performs language detection and translation using advanced AI models	Gemini 2.5 (Google Generative

			AI API)
4	Prompt Engineering Layer	Designs structured prompts for accurate language detection and translation	Custom Prompt Templates (Python)
5	Response Processing	Extracts translated text from API response and formats output clearly for users	Python
6	Deployment Environment	Local execution or cloud deployment of the Streamlit application	Streamlit Cloud / Local Deployment
7	Future Enhancement	Add user authentication, translation history, downloadable files, and multilingual UI support	Database Integration (Future Scope)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The application is built using open-source frameworks and libraries that support rapid development and integration with AI services. Streamlit is used for building the interactive web interface, and Python serves as the core programming language for backend logic and API integration.	Python, Streamlit
2	Security Implementations	The application secures API access using environment variables to store sensitive credentials such as the Gemini API key. No hardcoded keys are used. HTTPS-based secure API communication ensures encrypted data transfer between the application and Google Gemini servers.	Environment Variables (.env), HTTPS Encryption, Google API Security Standards
3	Scalable Architecture	The architecture follows a modular layered approach consisting of Presentation Layer (Streamlit UI), Application Layer (Python logic), and AI Service Layer (Gemini API). This separation ensures easy scalability and future migration to microservices or cloud deployment.	3-Tier Logical Architecture, Google Gemini API
4	Availability	The application can be deployed on Streamlit Cloud or any cloud platform to ensure continuous availability. Since translation processing is handled by Google's cloud-based Gemini infrastructure, high availability is ensured by Google's distributed servers.	Streamlit Cloud, Google Cloud Infrastructure
5	Performance	The system is designed for fast response time using lightweight Streamlit UI and optimized API calls. Gemini 2.5 Flash model ensures low latency translation responses. The application supports multiple requests per session efficiently.	Gemini 2.5 Flash Model, Streamlit Runtime